

Laboratory work 1. ERD Diagram — International Airport Database

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Description

The International Airport database supports the core operations of an airport: managing airports and airlines, scheduling flights, processing passenger bookings, issuing boarding passes, and tracking baggage and security checks. Passengers make bookings for flights; each booking may later be modified, generates a boarding pass, and may include registered baggage. Baggage is checked before departure, and every passenger passes a security check before boarding. The diagram below models these entities, their attributes, and the relationships that connect them.

1. Airports: airport_id (PK), airport_name, country, state, city, created_at, updated_at.
2. Airlines: airline_id (PK), airline_code, name, country, created_at, updated_at.
3. Flights: flight_id (PK), departing_gate, arriving_gate, scheduled_departure_time, scheduled_arrival_time, actual_departure_time, actual_arrival_time, airline_id (FK), departure_airport_id (FK), arrival_airport_id (FK), created_at, updated_at.
4. Passengers: passenger_id (PK), first_name, last_name, gender, date_of_birth, country_of_citizenship, country_of_residence, passport_number, created_at, updated_at.
5. Bookings: booking_id (PK), flight_id (FK), passenger_id (FK), status, booking_platform, ticket_price, created_at, updated_at.
6. Booking_Changes: change_id (PK), booking_id (FK), change_description, created_at, updated_at.
7. Boarding_Passes: boarding_pass_id (PK), booking_id (FK), seat, boarding_time, created_at, updated_at.
8. Baggage: baggage_id (PK), booking_id (FK), weight_in_kg, created_at, updated_at.
9. Baggage_Checks: baggage_checking_id (PK), booking_id (FK), passenger_id (FK), check_result, created_at, updated_at.
10. Security_Checks: security_check_id (PK), passenger_id (FK), check_result, created_at, updated_at.

Relationships & Cardinality Constraints

- **Airlines — Flights:** one-to-many (1:N). One airline operates many flights.
- **Airports — Flights:** one-to-many (1:N). One airport serves as the departure or arrival point for many flights.
- **Passengers — Bookings:** one-to-many (1:N). One passenger can create multiple bookings.
- **Flights — Bookings:** one-to-many (1:N). One flight contains many bookings.
- **Bookings — Booking_Changes:** one-to-many (1:N). One booking can have multiple change records.
- **Bookings — Boarding_Passes:** one-to-one (1:1). Each booking generates exactly one boarding pass.
- **Bookings — Baggage:** one-to-many (1:N). One booking can include multiple baggage items.
- **Bookings — Baggage_Checks:** one-to-many (1:N). One booking can have multiple baggage checks.
- **Passengers — Baggage_Checks:** one-to-many (1:N). One passenger can have multiple baggage checks recorded.
- **Passengers — Security_Checks:** one-to-many (1:N). A passenger undergoes one or more security checks.

